

BE Fourth Year
Computer Engineering
PROJECT SYNOPSIS
ON

**FocusGuard: Continuous Attention
Monitoring in Video Conferencing using
MediaPipe, LDTN and CNN**

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Abstract:

The rapid adoption of online learning and remote work has significantly increased the use of video conferencing platforms such as Zoom, Microsoft Teams, and Google Meet. However, instructors and meeting hosts face difficulties in continuously monitoring participants' attention and engagement due to the lack of physical interaction. Existing approaches primarily rely on eye gaze, facial expressions, or head pose independently, often resulting in inaccurate attention estimation and poor temporal understanding of user behavior. This paper proposes a **Continuous Attention Monitoring System** that integrates **MediaPipe Face Mesh**, **Long Dependency Temporal Network (LDTN)**, and **Convolutional Neural Networks (CNN)** to perform real-time attention analysis from webcam video streams. MediaPipe extracts facial landmarks, eye movements, head pose, and blink information, while CNN performs spatial feature extraction and LDTN captures temporal attention patterns over continuous video sequences. Based on these features, the proposed model classifies participant attention levels into attentive, distracted, and inactive states. The system further generates continuous attention scores, real-time alerts, and analytics for instructors or meeting hosts. The proposed framework aims to provide a lightweight, scalable, and privacy-preserving solution that operates using only a standard webcam without requiring specialized eye-tracking hardware.

Keywords :

Attention Monitoring, Video Conferencing, Student Engagement, MediaPipe Face Mesh, CNN, Long Dependency Temporal Network (LDTN), Eye Gaze Estimation, Facial Landmark Detection, Head Pose Estimation, Deep Learning, Human-Computer Interaction, Computer Vision.

Introduction:

The rapid growth of online education, hybrid classrooms, remote work, and virtual collaboration has transformed video conferencing into an essential communication medium. Platforms such as Zoom, Microsoft Teams, and Google Meet are now widely used for academic, corporate, and professional interactions. Although these platforms provide efficient communication, they lack mechanisms for automatically assessing whether participants remain attentive throughout a session.

In conventional classrooms, instructors can evaluate students' attention through eye contact, facial expressions, and body language. However, such cues become difficult to interpret in virtual environments, especially when multiple participants are present. Similarly, meeting organizers cannot accurately determine whether employees actively participate or become distracted during long meetings.

Recent research has explored student engagement detection using computer vision techniques including eye gaze estimation, head pose analysis, blink detection, facial expression recognition, and deep learning-based classification. While these approaches have achieved promising results, many rely on single-frame analysis or individual behavioral features, limiting their ability to capture continuous attention variations over time. Additionally, several existing methods require specialized eye-tracking devices or wearable sensors, increasing deployment cost and reducing scalability.

To address these limitations, this work proposes a Continuous Attention Monitoring System that combines MediaPipe Face Mesh, Convolutional Neural Networks (CNN), and a Long Dependency Temporal Network (LDTN). MediaPipe provides robust facial landmark extraction in real time using only a standard webcam. CNN extracts discriminative spatial features from facial images, whereas LDTN models temporal dependencies across consecutive frames to improve attention recognition. The proposed framework continuously estimates participant attention levels and generates attention scores, alerts, and analytics for meeting hosts while maintaining low computational complexity suitable for real-time deployment.

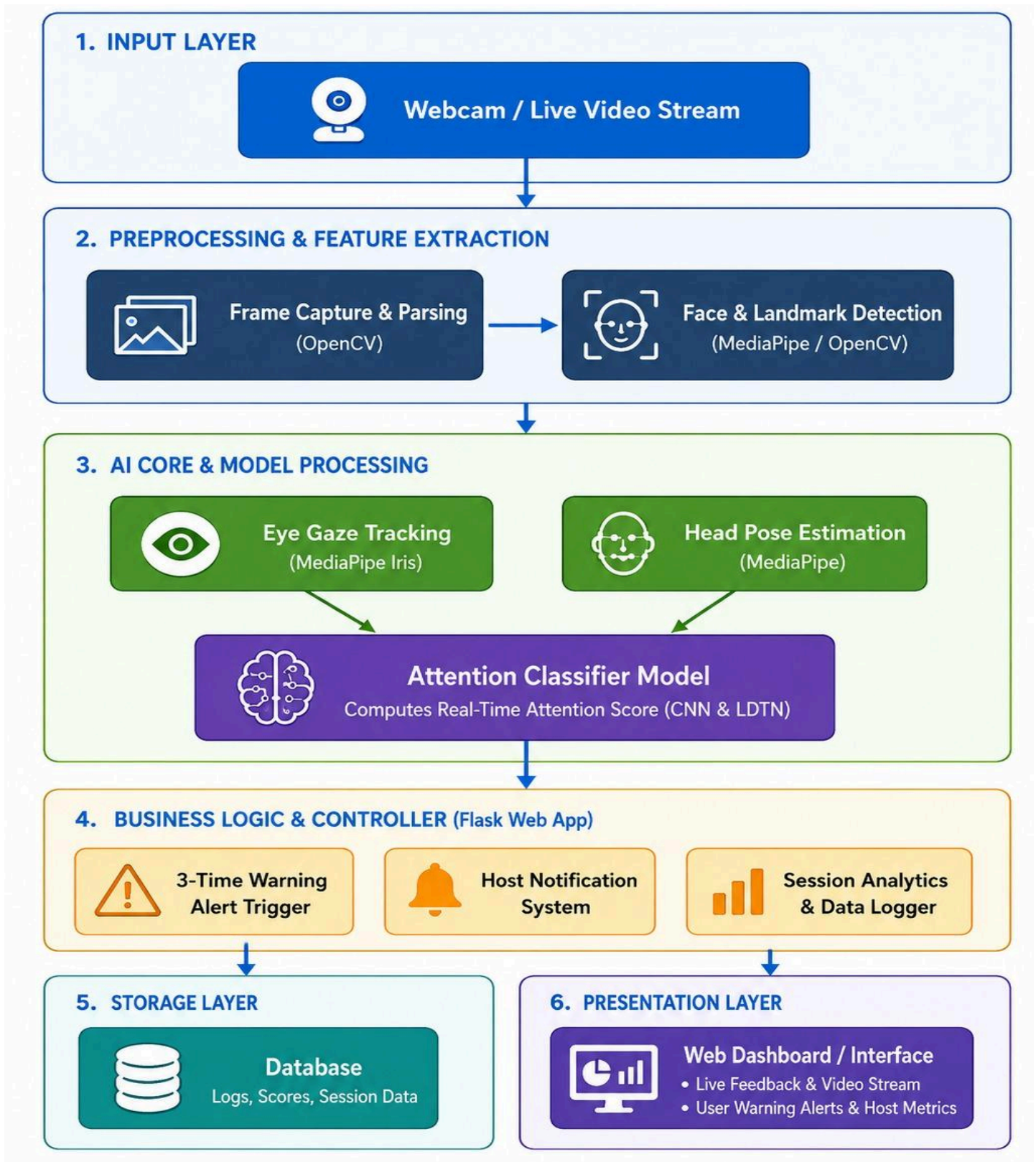


Fig 1: System Architecture
(Source : <https://openai.com/>)

Literature Survey

Table 1 : Literature Survey

Sr. No.	Year	Author(s)	Title	Methodology	Advantages	Limitations
1	2019	Liu Jigang et al.	Free-Head Appearance-Based Eye Gaze Estimation on Mobile Devices	Two-stage CNN (GazeEstimator) trained on 300W-LP and GazeCapture datasets	Robust gaze estimation under occlusion and partial face visibility; no specialized hardware	Focuses only on gaze estimation; does not perform attention or engagement classification
2	2022	Snigdha Das et al.	Stungage: Continuous Monitoring of Student Engagement	Context-aware engagement monitoring using gaze, presentation context, and cognitive cues	Supports productive multitasking and preserves user privacy through local processing	Does not utilize detailed facial landmarks or temporal deep learning models
3	2023	(CNN-MobileNet Paper Authors)	Student Engagement Detection using Deep CNN-based Learning Approach	CNN and MobileNet for engagement classification	Lightweight model with high classification accuracy (93%); suitable for real-time deployment	Requires manually labeled image dataset; frame-based analysis only
4	2024	Shreeya Patil et al.	Student's Attention and Alertness Monitoring System using AI	Face Detection + Facial Landmarks + Eye Aspect Ratio (EAR)	Simple real-time attention monitoring using webcam; easy implementation	Performance affected by lighting conditions; limited behavioral features
5	2024	Jie Wang et al.	Fusing YOLOv5s–MediaPipe–HRV to Classify Engagement in E-learning	YOLOv5s + MediaPipe + HRV + LSTM	Combines behavioral and physiological cues; robust multimodal engagement detection	Requires wearable HRV sensors; increases deployment complexity and cost
6	2024	Jayasankar Santhosh et al.	Gaze-Driven Adaptive Learning System with ChatGPT-Generated Summaries	Gaze tracking + InceptionTime + Transformer + ChatGPT	Adaptive learning improves engagement and comprehension through personalized feedback	Focuses on adaptive learning rather than continuous attention monitoring
7	2025	M. Vijay Kumar et al.	Deep Learning-based Student Engagement Detection using CNN, MobileNet, ResNet50, VGG16	Transfer Learning with CNN, MobileNet, ResNet50 and VGG16	MobileNet provides excellent balance between accuracy (92.5%) and efficiency	Small dataset; ResNet50 exhibits overfitting; lacks temporal modeling

Sr. No.	Year	Author(s)	Title	Methodology	Advantages	Limitations
9	2025	Jie Wang et al.	Video-based Real-Time Monitoring of Engagement in E-learning using MediaPipe through Multi-feature Analysis	MediaPipe + Head Pose + Blink + Gaze + Smile + XGBoost	Multi-feature analysis improves robustness; lightweight and edge-device friendly	Does not exploit temporal deep learning for continuous attention prediction
10	2025	Jie Zhao et al.	3D Facial Landmarks Video Analytics for Student Engagement	MediaPipe Face Mesh + 3D CEAR	Eliminates specialized eye trackers; 3D landmarks improve engagement estimation	Limited to CEAR-based attention assessment; fewer behavioral cues
11	2025	Riju Das et al.	Optimizing Student Engagement Detection using Facial and Behavioral Features	OpenFace + MediaPipe + Action Units + EfficientNet + XGBoost	Multimodal facial and behavioral features significantly improve engagement recognition	Deep learning models remain computationally expensive; moderate accuracy
12	2025	Nan Xie et al.	MSC-Trans: A Multi-Feature-Fusion Network with Encoding Structure for Student Engagement Detecting	CNN + ResNet + Transformer + Temporal Feature Fusion	State-of-the-art temporal engagement tracking with flexible multimodal fusion	Transformer-based architecture requires high computational resources
13	2025	Ali Abedi et al.	Improving State-of-the-Art in Detecting Student Engagement with ResNet and TCN Hybrid Network	ResNet + Temporal Convolutional Network (TCN)	Joint spatial-temporal learning improves engagement recognition accuracy	High computational cost limits deployment on low-resource devices
14	2025	Erick Noboa et al.	Real-time Gaze Estimation via Face Mesh and Machine Learning: A Case Study	MediaPipe Face Mesh + Support Vector Regression (SVR) + Head Pose	Low-cost webcam-based gaze estimation; no dedicated eye tracker required	Requires user-specific calibration before use

Sr. No.	Year	Author(s)	Title	Methodology	Advantages	Limitations
15	2026	Rasmi Ranjan Khansama et al.	Capturing Short- and Long-Term Temporal Dependencies Using Bahdanau-Enhanced Fused Attention Model for Financial Data—An Explainable AI Approach	BiGRU + Top-k Sparse Attention + Global Attention + Bahdanau Attention + XAI (SHAP, Integrated Gradients, Attention Visualization)	Captures both short- and long-term temporal dependencies effectively; explainable predictions through XAI; strong generalization across complex datasets.	Designed for financial time-series forecasting rather than computer vision; computational complexity is higher due to multiple attention modules; requires large sequential datasets.
16	2025	Congbing He et al.	CDTDNet: A Neural Network for Capturing Deep Temporal Dependencies in Time Series	Cell State Capture Recurrent Unit (CSCRU) + Temporal Convolutional Network (TCN) + Temporal Attention + Coupled Attention + AutoEncoder	Novel CSCRU captures forgotten temporal information; integrates shallow and deep temporal dependencies; robust performance across multiple datasets.	Proposed for generic time-series forecasting rather than video attention analysis; deeper architecture increases training complexity; limited validation on real-time vision applications.
17	2024	Qi Xiong et al.	Modeling Temporal Dependencies within the Target for Long-Term Time Series Forecasting	TDAlign framework with adaptive temporal dependency alignment loss; plug-and-play enhancement for existing forecasting models	Can improve existing forecasting models without modifying architecture; negligible computational overhead; consistently improves forecasting accuracy.	Focuses only on temporal dependency optimization for forecasting; not suitable as a standalone attention detection model; designed for numerical sequence prediction instead of image/video data.
18	2014	Nicolas Boulanger-Lewandowski et al.	Exploiting Long-Term Temporal Dependencies in NMF Using Recurrent Neural Networks with Application to Source Separation	Recurrent Neural Network (RNN) integrated with Non-negative Matrix Factorization (NMF) and RNN-RBM	One of the earliest works exploiting long-term temporal dependencies using RNNs; improves feature extraction by modeling event co-occurrence; enhances audio source separation.	Designed specifically for audio signal processing; high computational cost due to Backpropagation Through Time (BPTT); not applicable directly to computer vision or engagement recognition.

Related work:

Automatic attention monitoring has gained significant interest in the fields of computer vision and educational technology. Existing research primarily focuses on estimating user engagement through facial landmarks, eye gaze, head pose, blink detection, and deep learning techniques.

Several studies have employed Convolutional Neural Networks (CNNs) for engagement classification, achieving promising accuracy in recognizing attentive and inattentive participants. However, most CNN-based methods analyze individual frames and fail to capture temporal variations in user behavior.

Recent approaches utilize MediaPipe Face Mesh for efficient extraction of facial landmarks, enabling real-time analysis of gaze direction, head pose, and facial expressions using standard webcams. Temporal models such as LSTM, Temporal Convolutional Networks (TCN), and Transformer-based architectures have further improved engagement recognition by learning behavioral patterns across video sequences.

Despite these advancements, many existing systems rely on single-feature analysis, computationally intensive models, or specialized eye-tracking hardware. Therefore, there remains a need for a lightweight and real-time framework that combines robust facial feature extraction with temporal learning. The proposed system addresses these limitations by integrating MediaPipe Face Mesh, CNN, and Long Dependency Temporal Network (LDTN) for continuous attention monitoring in video conferencing environments.

Problem Statement :

The widespread adoption of online education and remote collaboration has increased the need for automated systems capable of continuously monitoring participant attention during video conferencing sessions. Existing platforms do not provide reliable mechanisms for evaluating whether participants remain attentive, distracted, or inactive throughout a meeting. Most current attention monitoring approaches depend on individual cues such as eye gaze or facial expressions, analyze isolated frames rather than continuous video streams, or require expensive eye-tracking devices and wearable sensors.

Therefore, there is a need for a lightweight, real-time, and hardware-independent system capable of continuously analyzing participant attention using multiple facial and behavioral features extracted from a standard webcam while providing meaningful attention analytics and alerts to instructors or meeting hosts.

Objectives of the Project

- To develop a real-time attention monitoring system for video conferencing environments.
- To extract facial landmarks, eye gaze, blink rate, and head pose using MediaPipe Face Mesh.
- To classify participant attention into attentive, distracted, and inactive states using CNN and LDTN.
- To continuously estimate participant attention scores throughout the meeting.
- To generate real-time alerts for prolonged inattentiveness.
- To provide an analytics dashboard for instructors or meeting hosts.
- To achieve high accuracy while maintaining low computational complexity using only a standard webcam.
- To eliminate the requirement for specialized eye-tracking hardware or wearable devices

Software Requirements

Category	Requirement
Operating System	Windows 10/11 or Ubuntu 22.04+
Programming Language	Python 3.10+
IDE	Visual Studio Code / PyCharm
Deep Learning Framework	TensorFlow 2.x or PyTorch 2.x
Computer Vision Library	OpenCV
Facial Landmark Detection	Google MediaPipe Face Mesh
Machine Learning	Scikit-learn
Database	SQLite / PostgreSQL
Backend Framework	Flask or FastAPI
Frontend	HTML, CSS, JavaScript (or React.js)
Version Control	Git & GitHub

Date:

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